Static Equipment Mechanical Integrity & Corrosion Manual

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Department:	Inspection & Integrity	**Site Location:**	Jamnagar Refinery Complex (JRC)

1. Scope and Authority

Governs statutory pressure vessels (V-201, V-220, V-501), atmospheric crude columns, heat exchangers (E-301), and storage tanks. Aligned with API 510, API 570, and ASME Section VIII Division 1.

2. Corrosion Damage Mechanisms

- Naphthenic Acid Corrosion (NAC): Severe metal attack between 220 °C and 380 °C in crude furnace outlet and transfer lines.
- Ammonium Chloride Salt Deposition: Salt precipitation in CDU column overhead causing under-deposit pitting.
- High Temperature Hydrogen Attack (HTHA): Governed by Nelson curves for HCU-1 reactor circuits.

3. Non-Destructive Examination (NDT) Schedule

Ultrasonic thickness (UT) grid measurements must be conducted annually on high-risk nozzles. Flange face phased array ultrasonic testing (PAUT) required during turnarounds.